



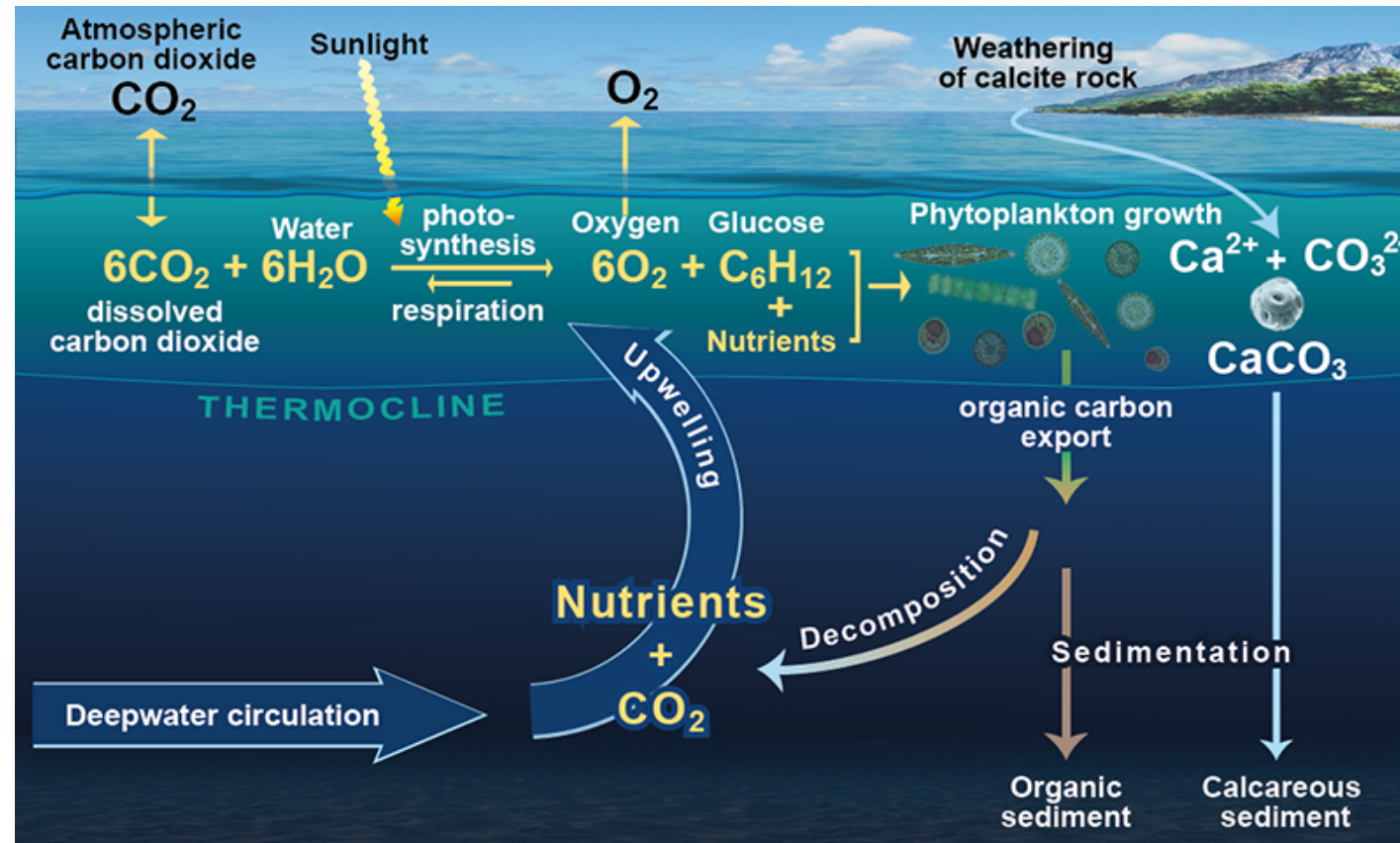
From lecture session 12: we learned that the ocean is capable of dissolving gases (like O_2 , N_2 , and CO_2) and that they are essential to life in the ocean (and on land). We also learned how CO_2 impacts the delicate pH balance of seawater.

Lecture Session 13

1. Cycles/processes that govern O_2 , N_2 , and CO_2 concentrations
2. Measuring their concentrations
3. Global distribution of gases
4. Human impacts

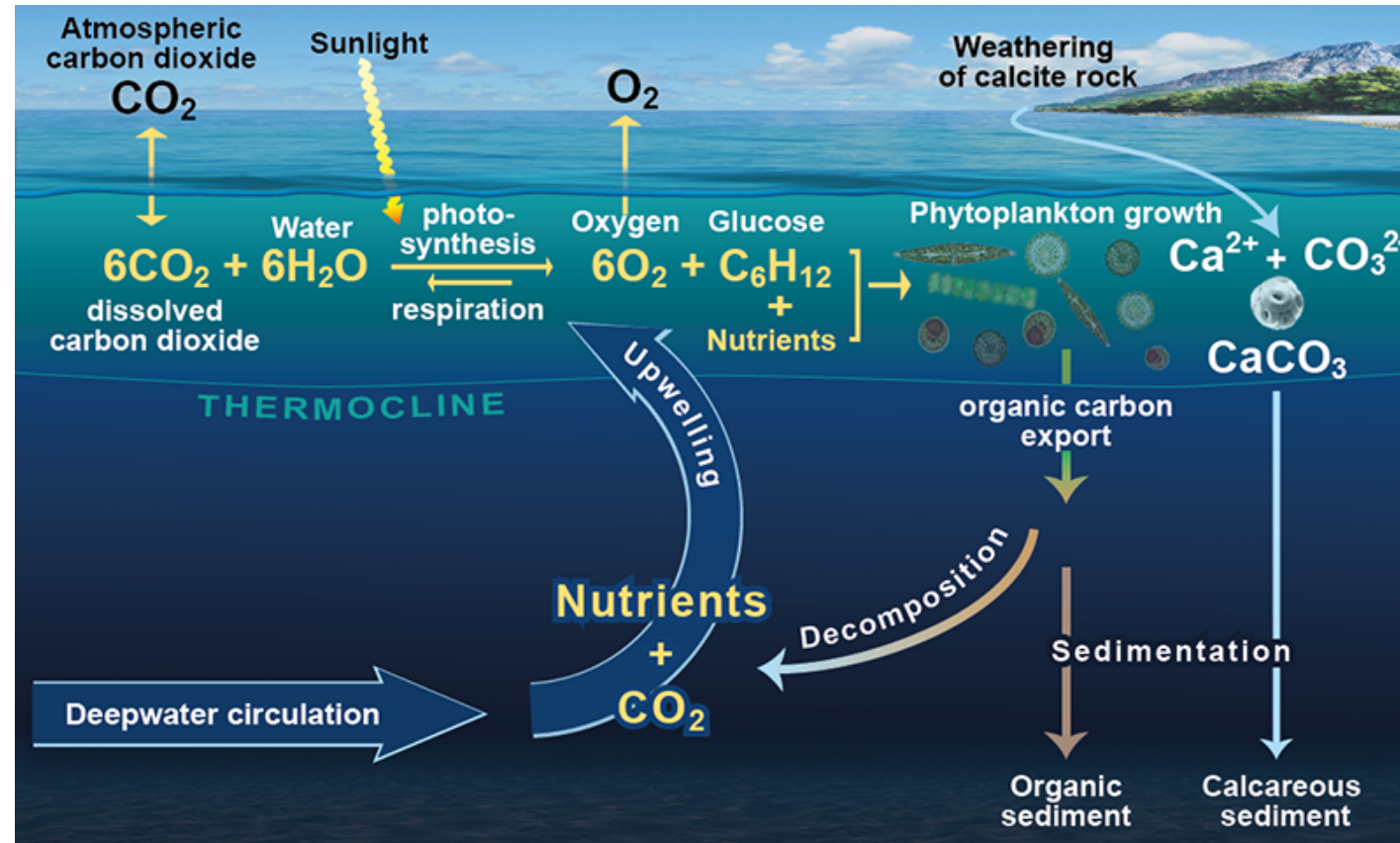
Thinking moment

Looking at the C cycle schematics, can you see why **Marine Biogeochemistry** is probably the most interdisciplinary of the ocean sciences?



Thinking moment

Processes governing the concentrations of these gases are intertwined!



Nitrogen - N₂

- Nitrogen is a nutrient that is one of the building blocks of life (like C and P)
 - Redfield ratio of nutrients in organic material ~ C:N:P = 106:16:1
- Among the different nutrients, N is the most limiting to life in the ocean (more than P, C, and other trace elements)
- N₂ gas (like in the atmosphere) is inert and unavailable to **most life forms**
- There are special kinds of bacteria and bacteria-like algae (aka nitrogen fixers) that convert N₂ into bio-available of N and promoting the so-called Nitrogen Cycle

(Nitrogen Fixation)

Nitrogen Cycle

Acronimys and important names to understand the schematics

- PON = Particulate Organic Nitrogen
- DON = Dissolved Organic Nitrogen
- Anammox = Anaerobic ammonium oxidation
- NH_4^+ = Ammonium
- NO_2^- = Nitrite
- NO_3^- = Nitrate

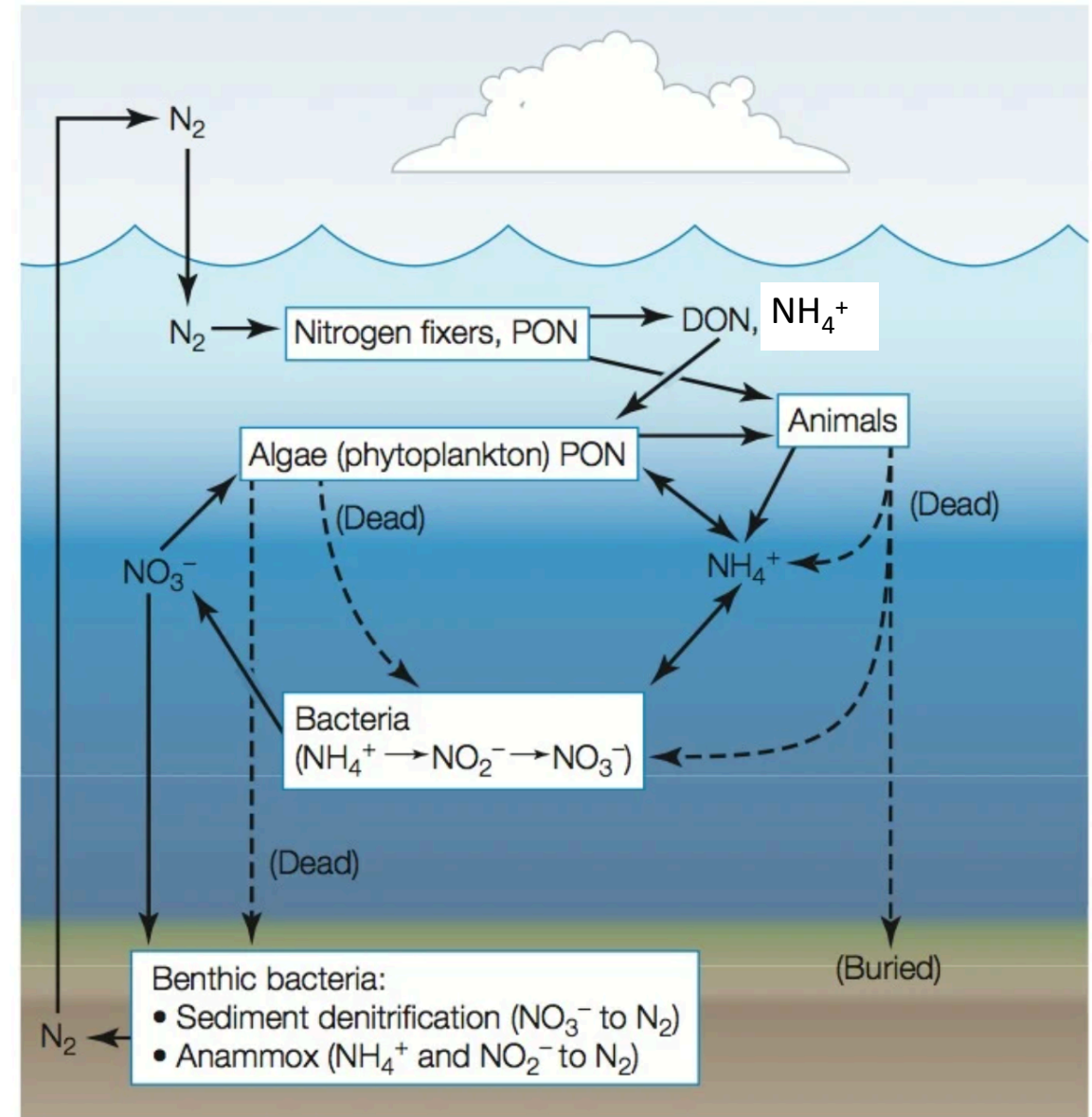


FIGURE 7.5 Flow diagram of the major processes involved in the marine nitrogen cycle. See text for details.

Nitrogen Cycle

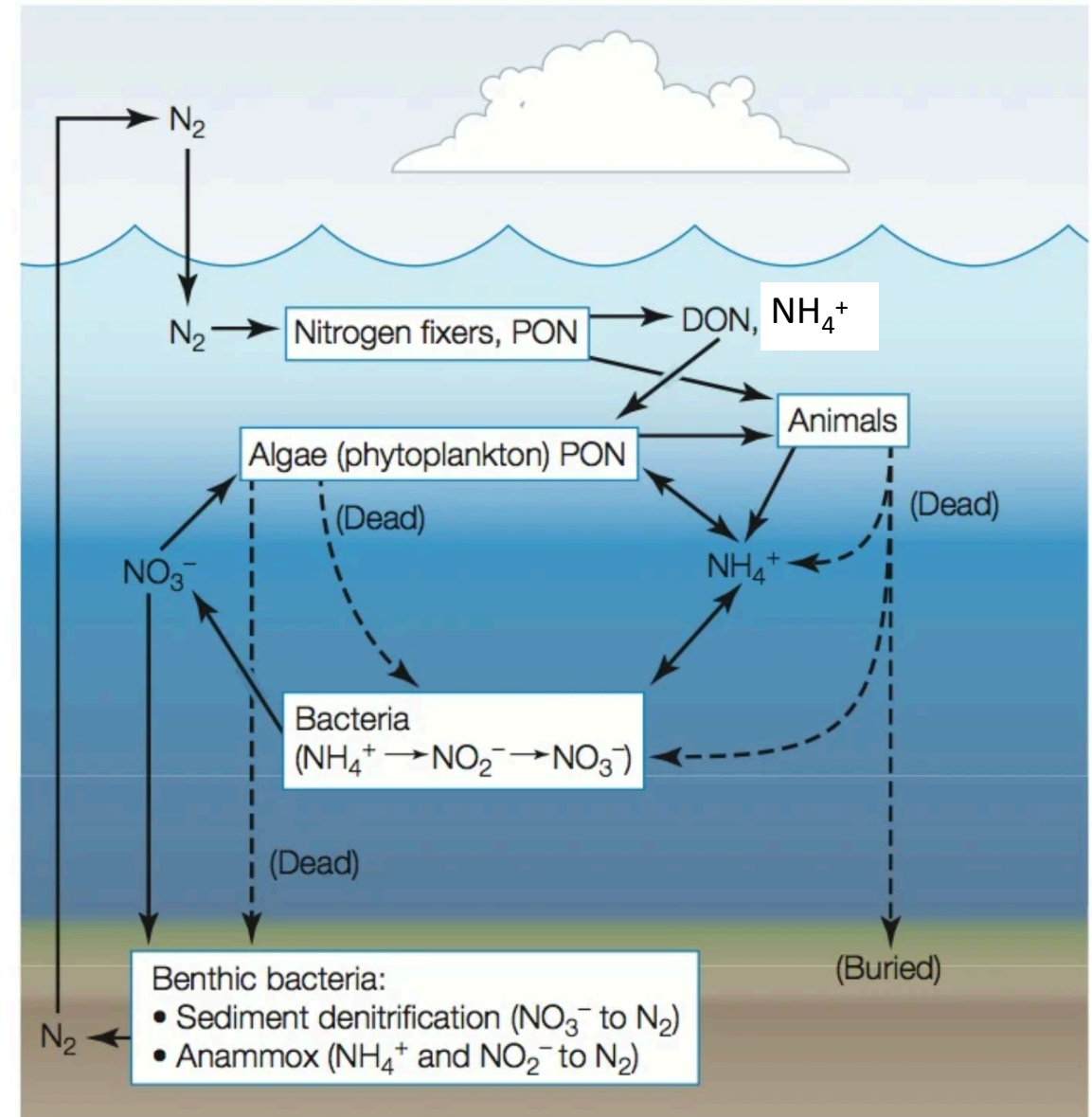


FIGURE 7.5 Flow diagram of the major processes involved in the marine nitrogen cycle. See text for details.

Nitrogen Cycle

Nitrogen Fixation

Diazotrophs can convert N_2 to PON, DON, and NH_4^+ via cellular activity

**Ex: Cyanobacteria
of the genus
Trichodesmium**

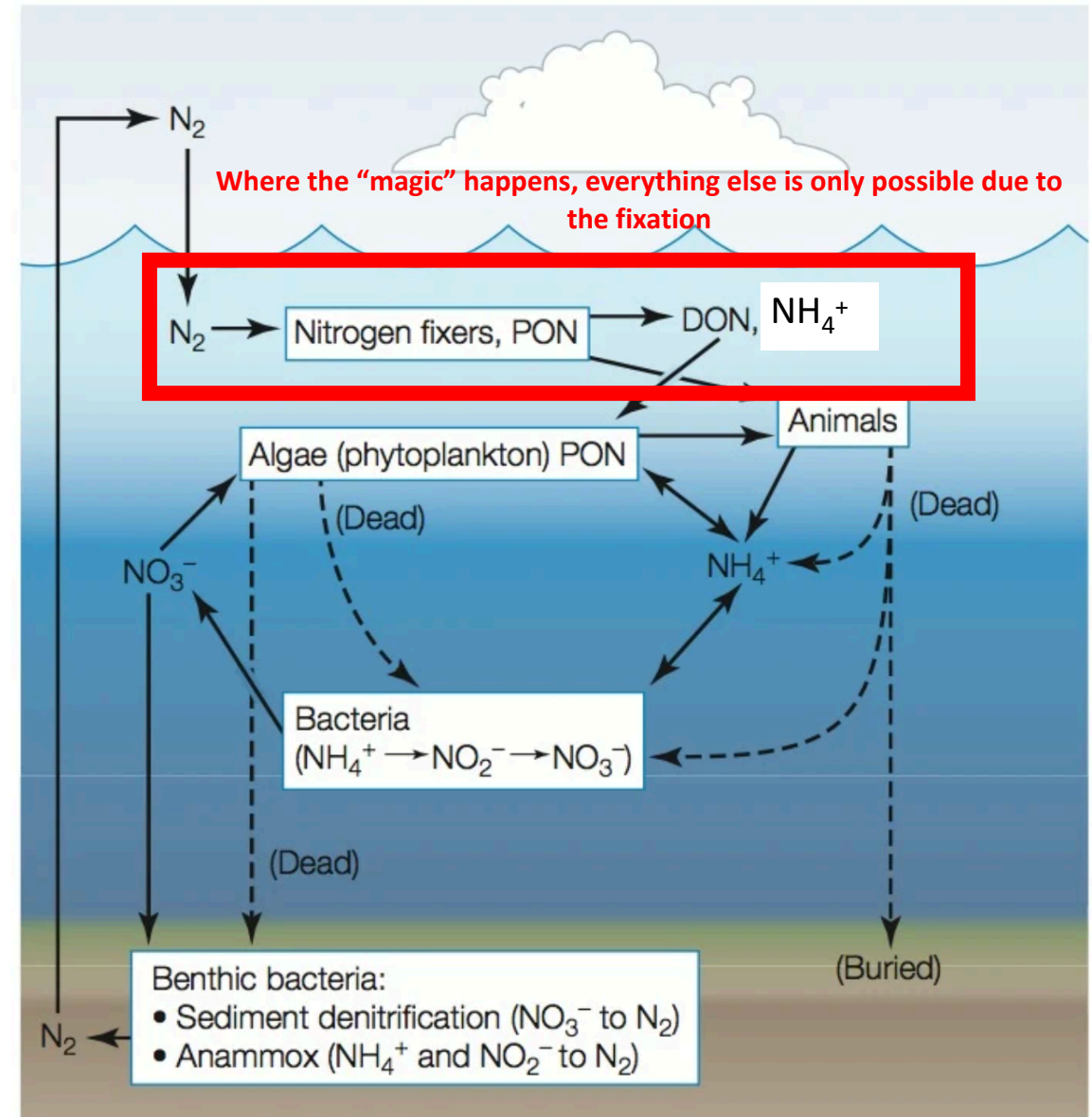


FIGURE 7.5 Flow diagram of the major processes involved in the marine nitrogen cycle. See text for details.

Nitrogen Cycle

After fixation ...

- Algae and Animals consume the different bio-available forms of N
- **In the dark:** Nitrification process (oxidation of NH_4^+)
- **At the bottom:** Denitrification and Anammox (the cycle closes)

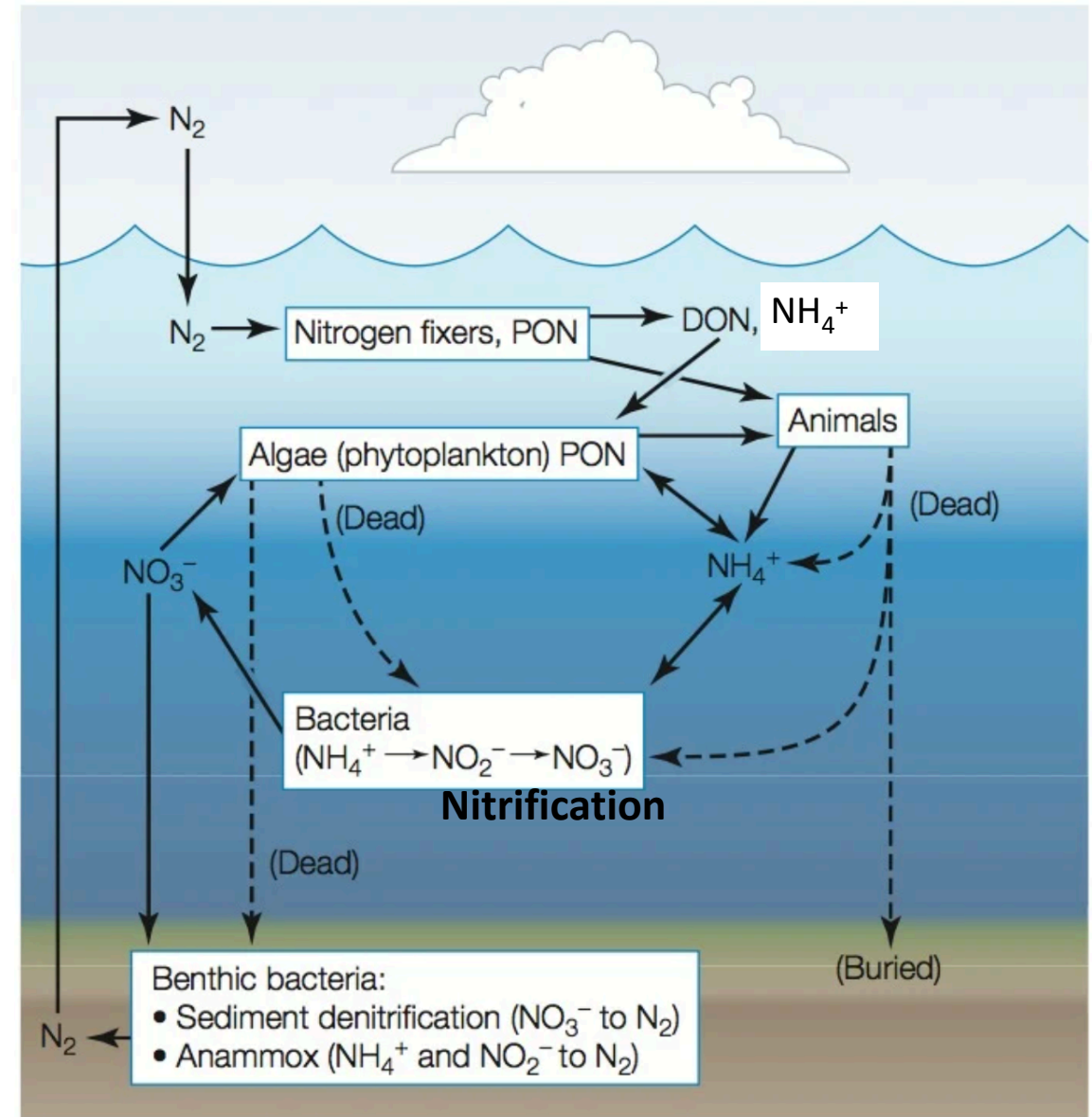
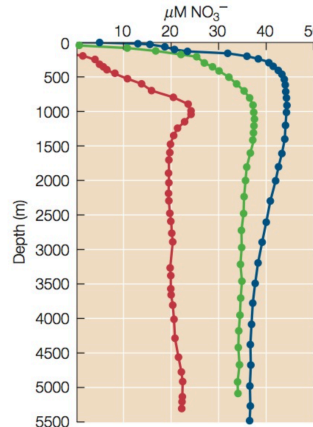


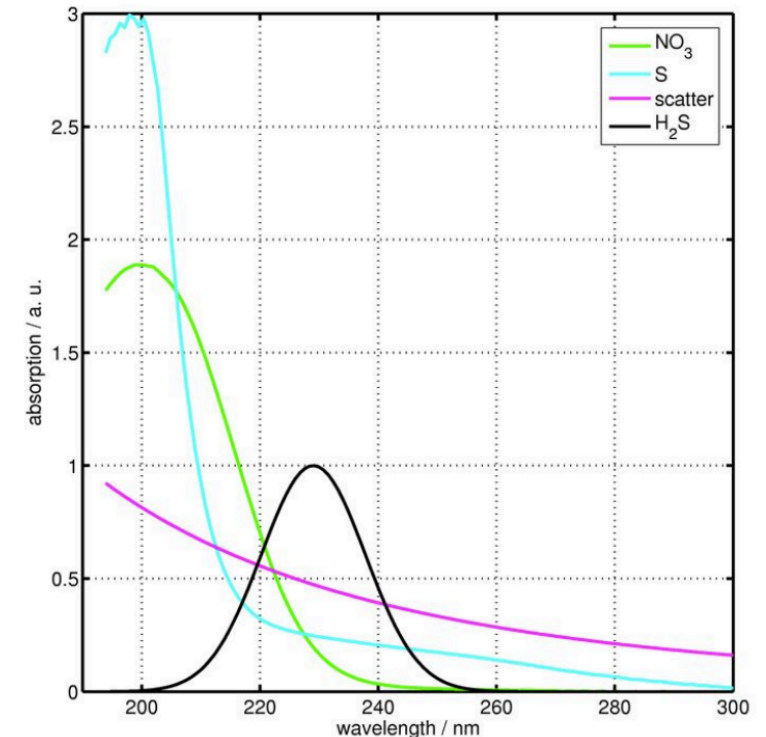
FIGURE 7.5 Flow diagram of the major processes involved in the marine nitrogen cycle. See text for details.

Measuring the different forms of Nitrogen

Traditionally, water samples are collected and N_2 , NH_4^+ , NO_2^- , and NO_3^- are analyzed in laboratory

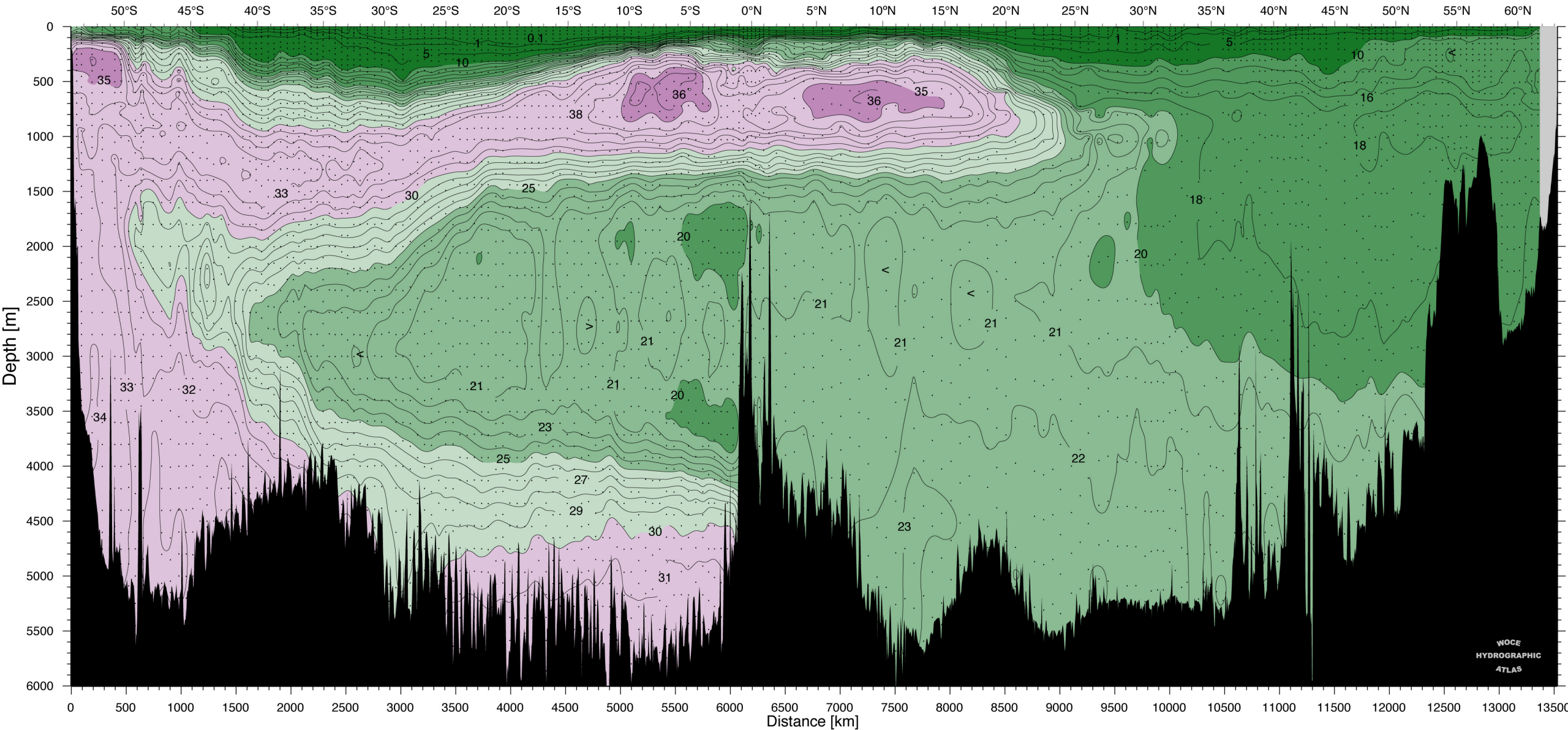
Nowadays, at least NO_3^- , can be measured optically

- Nitrate absorbs UV light with a characteristic spectrum that has a maximum at ~ 200 nm
- By shining UV light through the water and measuring the magnitude of the absorption spectrum, nitrate concentration can be calculated



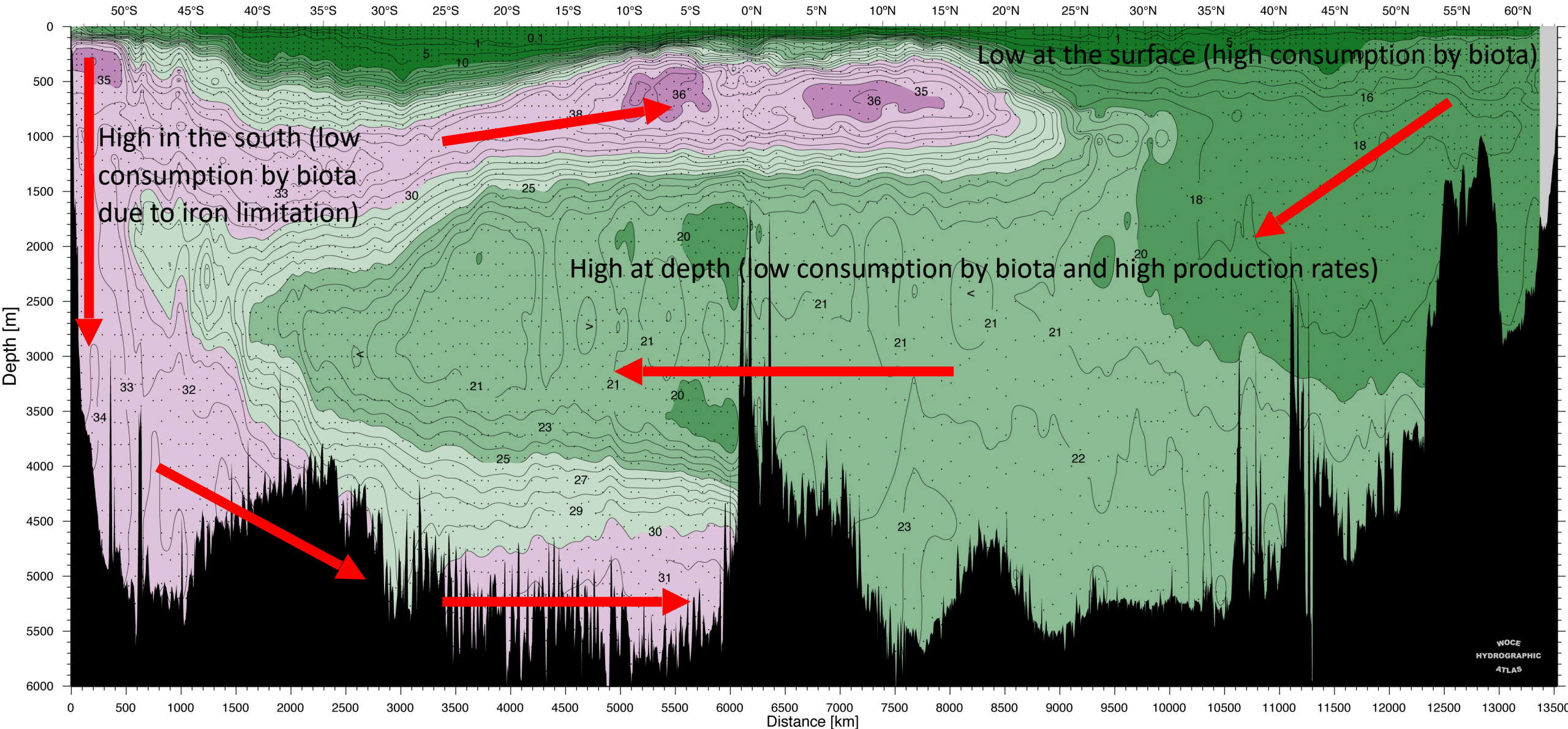
Nitrate (μ M/kg) distribution

What patterns do you see?
Why?



Nitrate (μ M/kg) distribution

Production, consumption, and ocean circulation are responsible



Human impacts

Eutrophication: When an environment becomes enriched with nutrients (ie nitrate), increasing the amount of plant and algae growth (typically in estuaries and coastal waters).



Causes

- Excess agricultural run-off (fertilizers and animal waste)
- Human sewage and industrial waste

Impacts

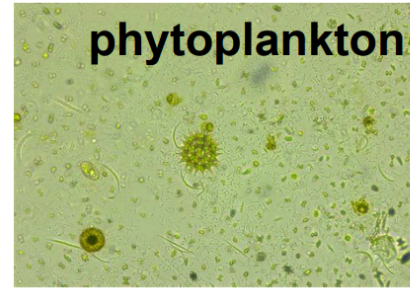
- Algal blooms can sometimes be toxic (Harmful Algal Blooms)
- Decay can lead to low O_2 conditions, fish die-off, and acidification (CO_2 release)

Oxygen - O₂

- Crucial for life
- Important tracer of ocean-atmosphere exchange and ocean circulation



Oxygen cycling



phytoplankton and seaweed



Photosynthesis

VS

Cellular Respiration

Photosynthesis



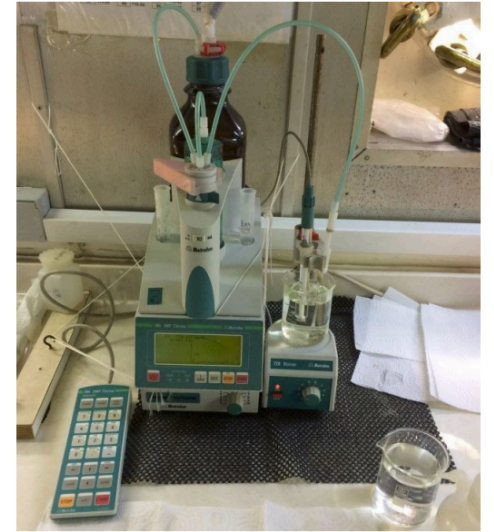
Respiration



Measuring Oxygen

Traditionally:

- Water samples are collected + titration method (aka Winkler method)
- Winkler method dates back to 1888, but the same principle is still applied today
- Still corresponds to the most accurate methodology



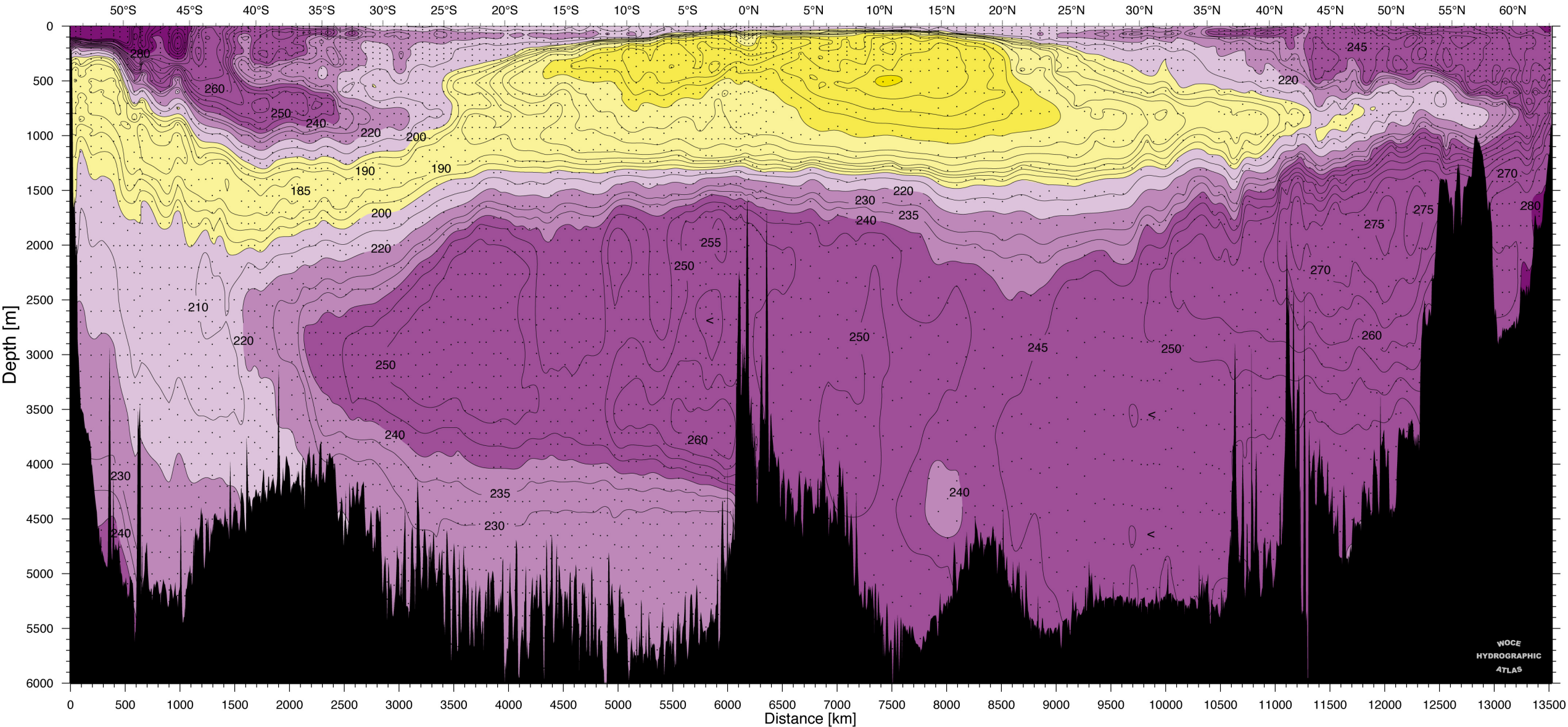
Modern methods involve the usage of optical sensors

- An oxygen-sensitive sensing foil that contains a substance that when excited with light (of certain wavelength) emits light with a different wavelength
- The amount of O_2 in the water determines the amount of “excitation” that occurs



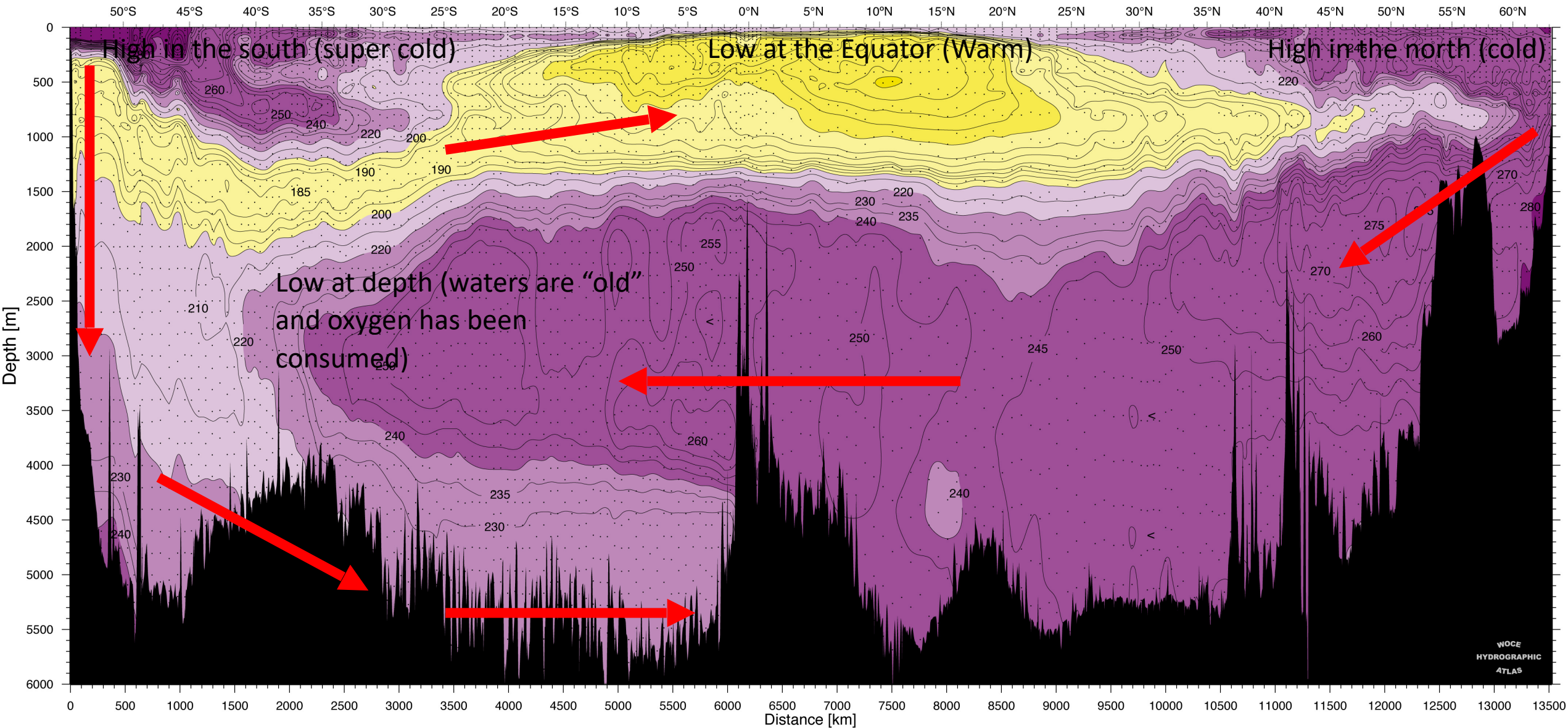
Oxygen ($\mu\text{M/kg}$) distribution

What patterns do you see (remember last class)?
Why?



Oxygen ($\mu\text{M/kg}$) distribution

Temperature, consumption, and ocean circulation are responsible



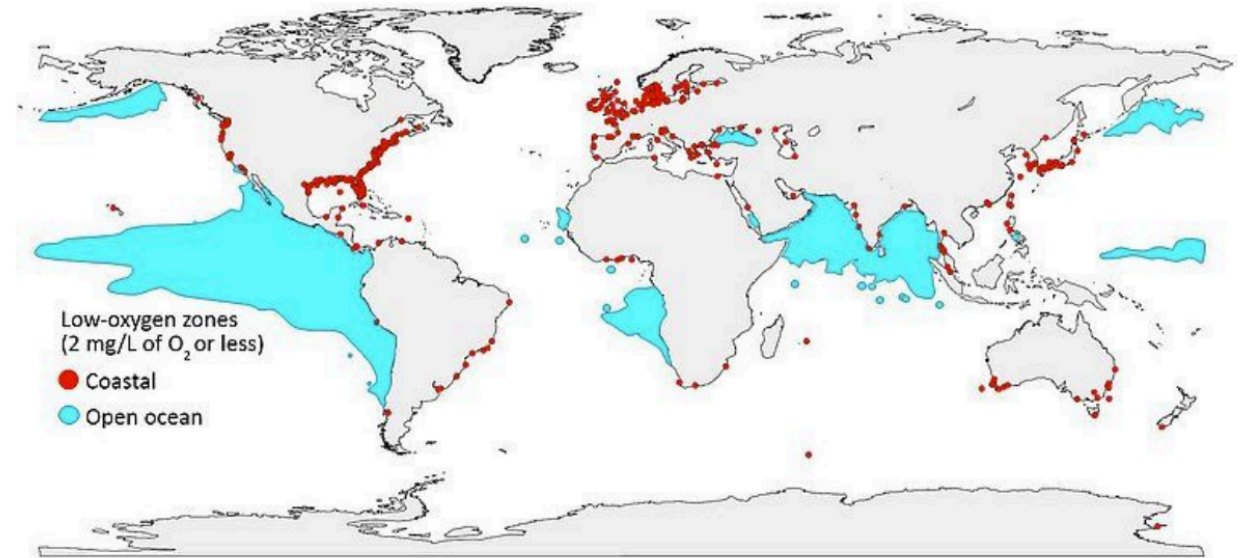
Oxygen Minimum Zones and “Dead Zones”

Hypoxia: a reduced level of oxygen in the water

Anoxia: no oxygen in the water

Human impacts

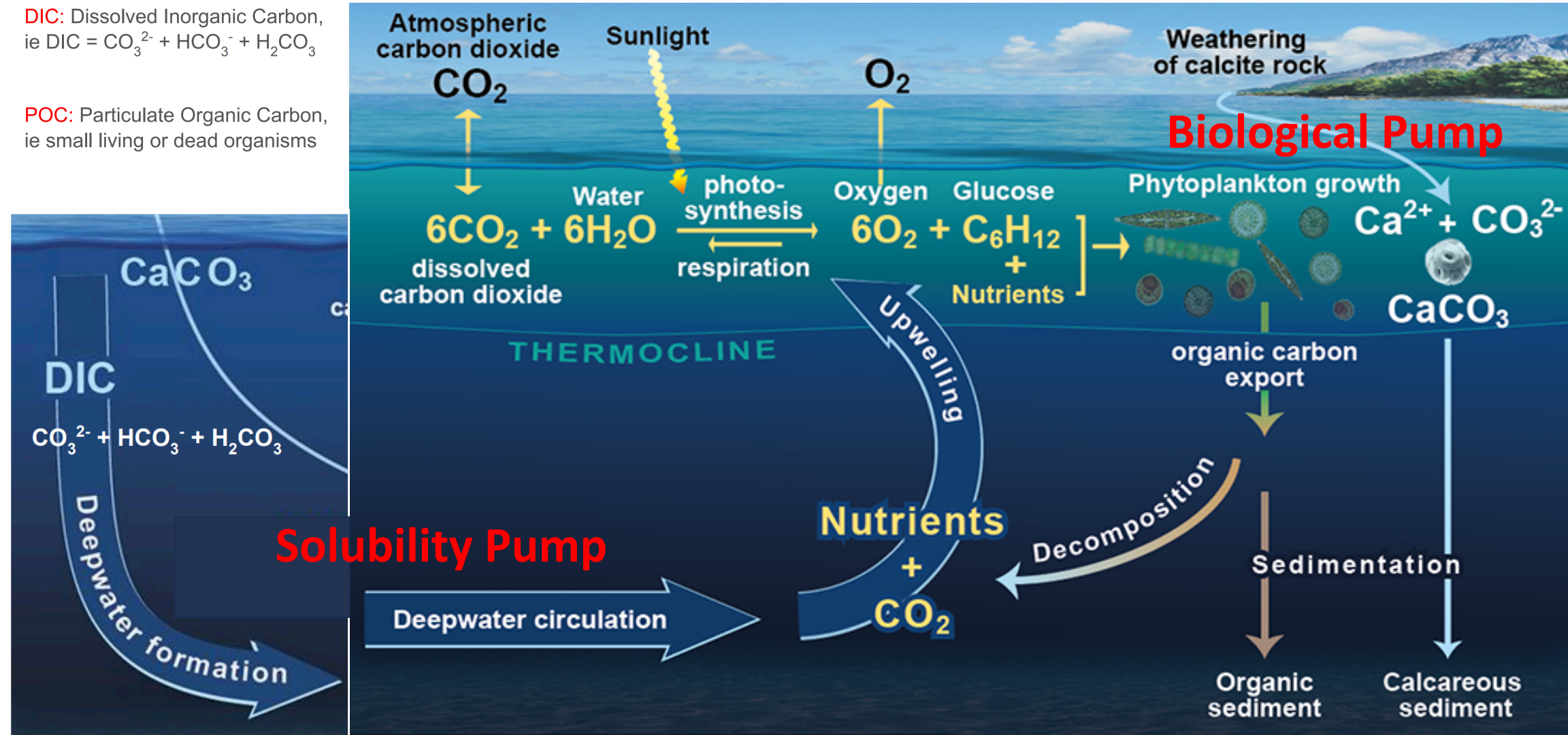
- Warming oceans → O_2 is less soluble
- Warming oceans → water is more stratified → harder for O_2 -rich waters to be mixed down
- Nutrient-rich pollution (eg. agricultural run-off) → algal blooms that rapidly decompose and use up all the O_2 through respiration, in addition to releasing CO_2



Carbon - CO₂, Dissolved Inorganic Carbon (DIC), Particulate Organic Carbon

DIC: Dissolved Inorganic Carbon,
ie $\text{DIC} = \text{CO}_3^{2-} + \text{HCO}_3^- + \text{H}_2\text{CO}_3$

POC: Particulate Organic Carbon,
ie small living or dead organisms



Measuring Carbon

DIC

- Using a coulometer, the amount of DIC in a sample can be determined by the amount of electricity used in electrolysis



POC

- Can be measured by means of combustion. A water sample is filtered and then the filter containing the POC mass is burnt; the amount of carbon released can be measured.



CO₂ (μM/kg) distribution

Any similarities to Nitrate and Oxygen patterns?

